Lecture 17

Curvature

Objectives. *Curvature measures the change in a vector transported around a small loop. We compute it from Christoffel symbols and from differential forms in an orthonormal frame. The sphere fixes the sign used for surface geometry and gravitation.*

By the end of this lecture you will be able to:

- Compute curvature with the output and direction indices specified.
- Prove tensoriality, the metric antisymmetries and the first Bianchi identity.
- Use Cartan's equations to find the curvature of a surface.
- Distinguish zero curvature from zero Christoffel symbols in one chart.

Unit 17A The curvature tensor and its contractions

17.1 The curvature tensor

Parallel transport compares tangent vectors along a path, as in Lecture 16.

Definition 17.1. Let M be a smooth manifold with an affine connection ∇. The *curvature operator* is defined on smooth fields by

$$R(X,Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

On the unit sphere, take the loop from the north pole $N = e_z$ to e_x, then to e_y, then back to N, along the shorter great-circle arcs. We orient the sphere by its outward normal. This loop is counter-clockwise when viewed from outside.

The vector initially equal to e_x reaches the vertex e_x as $-e_z$. It stays equal to $-e_z$ along the equator and returns to N as e_y. Along a great circle its tangent is parallel. A constant vector perpendicular to that circle's plane is also parallel, because its ambient derivative is zero. These observations verify the stated transports by tangent projection. The return vector has rotated by $+\pi/2$ in $T_N S^2$. The figure compares this with transport around a planar triangle.

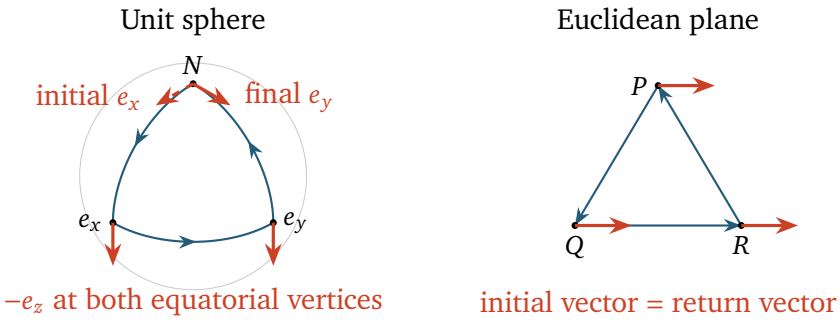


Figure 17.1: Parallel transport around the oriented octant sends e_x to e_y at N. The planar comparison returns its initial vector unchanged.

The octant has area $\int_0^{\pi/2}\int_0^{\pi/2}\sin\theta\,d\theta\,d\phi = \pi/2$. Lecture 18 relates the rotation to the integral of surface curvature.

For the Cartesian connection on $\mathbb{R}^n$, the components of $R(X,Y)Z$ are $XY(Z^k) - YX(Z^k) - [X,Y](Z^k) = 0$. The bracket correction removes the failure of the direction fields to commute.

Curvature measures the leading change under transport around a small loop. Choose local coordinates $(x, y, \ldots)$ about p. Traverse the coordinate square in the order $+x, +y, -x, -y$, with side $\varepsilon > 0$ and the other coordinates fixed. Write F_{xy} for the matrix of $R(\partial_x, \partial_y)$ at p in a chosen basis of T_pM. The parallel-transport matrix around this square satisfies

$$\mathrm{Hol}(\square_\varepsilon) = \mathbb{1} - \varepsilon^2 F_{xy} + O(\varepsilon^3).$$

The coordinate directions commute, so the square closes. Lecture 21 proves this small-square formula. The tensoriality proved next says that the curvature coefficient depends only on the vectors at p, independently of their extensions.

Theorem 17.2 (Tensoriality of curvature)**.** *Curvature is linear over $C^\infty(M)$ in all three arguments. It therefore gives a smooth tensor of type $(1,3)$.*

Proof. Real multilinearity is immediate from the connection rules. For $f \in C^\infty(M)$, the Leibniz rule and $[fX,Y] = f[X,Y] - Y(f)X$ give

$$R(fX,Y)Z = fR(X,Y)Z - Y(f)\nabla_X Z + Y(f)\nabla_X Z = fR(X,Y)Z.$$

Antisymmetry in X and Y gives linearity in Y. Expansion in the last slot gives

$$R(X,Y)(fZ) = fR(X,Y)Z + \big(XY(f) - YX(f) - [X,Y](f)\big)Z = fR(X,Y)Z.$$

The cross terms $Y(f)\nabla_X Z$ and $X(f)\nabla_Y Z$ cancel between the two second derivatives. By the cutoff and coordinate-expansion argument of §16.1, the value therefore depends only on the three vectors at the point. Smoothness follows from the local coefficients of the connection. ∎

Definition 17.3. The tensor of Theorem 17.2 is called the *Riemann curvature tensor*.

Write $\partial_\mu = \partial/\partial x^\mu$ in a chart. Greek indices here range from 1 to n. The index order is fixed by the action

$$R(\partial_\mu, \partial_\nu)\partial_\sigma = R^\rho{}_{\sigma\mu\nu}\partial_\rho.$$

Applying the connection product rule twice gives

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu\Gamma^\rho_{\nu\sigma} - \partial_\nu\Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}.$$

For example, applying the product rule gives

$$\nabla_{\partial_\mu}\nabla_{\partial_\nu}\partial_\sigma = (\partial_\mu\Gamma^\rho_{\nu\sigma} + \Gamma^\lambda_{\nu\sigma}\Gamma^\rho_{\mu\lambda})\partial_\rho.$$

Subtract the reversed expression. The bracket term vanishes because coordinate fields commute. Curvature is a tensor even though the Christoffel symbols are not. For the Euclidean plane, Lecture 16 gives the polar symbols $\Gamma^r_{\phi\phi} = -r$ and $\Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = 1/r$. For example, the curvature component is

$$R^r{}_{\phi r\phi} = \partial_r(-r) - (-r)(1/r) = -1 + 1 = 0.$$

All curvature components vanish because the tensor is zero in Cartesian coordinates.

Definition 17.4. A connection is *flat* when its curvature is zero.

Example 17.5 (The sphere in coordinates)**.** We compute a curvature component on the sphere of radius $a > 0$. Its spherical metric is

$$g = a^2\mathrm{d}\theta \otimes \mathrm{d}\theta + a^2\sin^2\theta\mathrm{d}\phi \otimes \mathrm{d}\phi, \qquad 0 < \theta < \pi.$$

The constant scale does not change Lecture 16's Christoffel symbols. Substituting them into the coordinate formula gives

$$\begin{aligned} R^\theta{}_{\phi\theta\phi} &= \partial_\theta(-\sin\theta\cos\theta) - (-\sin\theta\cos\theta)\cot\theta \\ &= (\sin^2\theta - \cos^2\theta) + \cos^2\theta = \sin^2\theta. \end{aligned}$$

Lowering the output with the metric gives $R_{\theta\phi\theta\phi} = a^2\sin^2\theta$. This positive component fixes the operator sign on the unit sphere.

17.2 Metric symmetries and the first Bianchi identity

Now let ∇ be the Levi-Civita connection of a pseudo-Riemannian metric. Lower the output index first: $R_{\rho\sigma\mu\nu} = g_{\rho\lambda}R^\lambda{}_{\sigma\mu\nu}$. For the proofs, abbreviate the four-field expression by $C(X,Y,Z,W) = g(R(X,Y)Z,W)$.

Proposition 17.6 (Curvature symmetries · *sketched; antisymmetries and first Bianchi proved*)**.** *The curvature satisfies*

$$\begin{aligned} C(X,Y,Z,W) &= -C(Y,X,Z,W) = -C(X,Y,W,Z), \\ R(X,Y)Z + R(Y,Z)X + R(Z,X)Y &= 0, \\ C(X,Y,Z,W) &= C(Z,W,X,Y). \end{aligned}$$

The middle equation is the first Bianchi identity.

Proof and sketch. The first antisymmetry follows from the operator definition. For the other one, apply $XY - YX - [X,Y]$ to $g(Z,W)$. It vanishes on this scalar function. Metric compatibility expands the result as

$$0 = g(R(X,Y)Z,W) + g(Z,R(X,Y)W),$$

with the two cross products of first derivatives cancelling. This proves antisymmetry in the last pair.

For Bianchi, sum cyclically over (X, Y, Z). The six second derivatives combine by zero torsion as follows:

$$\begin{aligned}\nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z + \nabla_Y\nabla_Z X - \nabla_Z\nabla_Y X + \nabla_Z\nabla_X Y - \nabla_X\nabla_Z Y \\ = \nabla_X(\nabla_Y Z - \nabla_Z Y) + \nabla_Y(\nabla_Z X - \nabla_X Z) + \nabla_Z(\nabla_X Y - \nabla_Y X) \\ = \sum_{\text{cyc}} \nabla_X [Y, Z].\end{aligned}$$

Relabel the bracket-direction terms to obtain

$$\sum_{\text{cyc}} R(X, Y)Z = \sum_{\text{cyc}} \big(\nabla_X [Y, Z] - \nabla_{[Y,Z]} X\big) = \sum_{\text{cyc}} [X, [Y, Z]] = 0.$$

The last equality is Jacobi's identity.

For pair exchange, first pair Bianchi with W:

$$C(X, Y, Z, W) + C(Y, Z, X, W) + C(Z, X, Y, W) = 0.$$

Write this scalar identity for four cyclic permutations of the four fields. Add the identities and use the two antisymmetries to cancel terms. The remaining equality gives pair exchange after renaming the fields. This sketches the algebraic step; see Lee, Proposition 7.4, pp. 121–123. ∎

In the lowered-output convention, these identities give $R_{abcd} = -R_{bacd} = -R_{abdc}$ and $R_{abcd} = R_{cdab}$. Our lowered indices correspond to the action $g(R(\partial_c, \partial_d)\partial_b, \partial_a)$.[1] Changing the operator sign negates curvature values. Changing slot order alone can also introduce a sign.

17.3 Ricci, scalar and sectional curvature

Definition 17.7. For a Riemannian metric and a tangent 2-plane $P \subset T_pM$, its *sectional curvature* is

$$K(P) = \frac{g(R(u, v)v, u)}{g(u, u)g(v, v) - g(u, v)^2}, \qquad P = \operatorname{span}\{u, v\},$$

where u and v are linearly independent.

The denominator is positive. On changing the basis of P, antisymmetry in both pairs multiplies the numerator by the square of the basis-change determinant. The Gram determinant in the denominator has the same factor. Thus the quotient is independent of the basis. For orthonormal u and v, it is $g(R(u, v)v, u)$. In indefinite signature the same quotient defines sectional curvature only for nondegenerate planes, whose Gram determinant is nonzero. For the sphere in Example 17.5, the Gram determinant is $a^4 \sin^2\theta$, so its tangent plane has $K = 1/a^2$.

Definition 17.8. The *Ricci tensor* is the trace

$$\operatorname{Ric}(X, Y) = \operatorname{tr}(Z \mapsto R(Z, Y)X).$$

Definition 17.9. The *scalar curvature* R_{sc} is the metric trace of Ric.

[1] Lee uses $\operatorname{Rm}_{\text{Lee}}(X, Y, Z, W) = C(X, Y, Z, W)$. Thus $R_{abcd} = \operatorname{Rm}_{\text{Lee}}(\partial_c, \partial_d, \partial_b, \partial_a)$, with the same operator sign.

In coordinates these traces are

$$\mathrm{Ric}_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}, \qquad R_{\mathrm{sc}} = g^{\mu\nu}\,\mathrm{Ric}_{\mu\nu}\,.$$

Pair exchange makes Ricci symmetric: in $\mathrm{Ric}_{ij} = g^{ab}R_{aibj}$, exchange the curvature pairs and then rename a and b. The symbol R_{sc} denotes scalar curvature, while a denotes the sphere radius.

For a Riemannian metric, let $(e_1, \dots, e_n)$ be an orthonormal basis at p. The trace definitions give

$$\mathrm{Ric}(e_1, e_1) = \sum_{i=2}^{n} g(R(e_i, e_1)e_1, e_i) = \sum_{i=2}^{n} K(\mathrm{span}\{e_1, e_i\}),$$
$$R_{\mathrm{sc}} = 2\sum_{i<j} K(\mathrm{span}\{e_i, e_j\}).$$

The omitted term in the first trace is zero by direction antisymmetry. Each plane occurs twice in the second trace, once in each order. Thus Ricci sums the plane curvatures through a unit direction. Scalar curvature sums over all orthogonal basis planes, with this factor of two. On a surface there is only one tangent 2-plane, so $\mathrm{Ric} = Kg$ and $R_{\mathrm{sc}} = 2K$. The first identity follows from the directional trace and polarisation. These unsigned sums require a positive metric. The metric traces remain the definitions in indefinite signature. Lecture 19 uses Ricci and scalar curvature in the gravitational field equation.

Curvature of a bi-invariant metric

The connection of Lecture 16 also gives curvature directly from Lie brackets.

Proposition 17.10 (Bi-invariant curvature). *For a bi-invariant pseudo-Riemannian metric on a Lie group and left-invariant fields X, Y and Z,*

$$R(X,Y)Z = -\frac{1}{4}[[X,Y],Z].$$

For a Riemannian metric and an orthonormal left-invariant pair X and Y,

$$K(\mathrm{span}\{X,Y\}) = \frac{1}{4}\|[X,Y]\|^2.$$

Proof. Lecture 16 gives $\nabla_X Y = \frac{1}{2}[X,Y]$ for left-invariant fields. All brackets below are again left-invariant, so

$$\begin{aligned} R(X,Y)Z &= \frac{1}{4}\big([X,[Y,Z]] - [Y,[X,Z]]\big) - \frac{1}{2}[[X,Y],Z] \\ &= \frac{1}{4}[[X,Y],Z] - \frac{1}{2}[[X,Y],Z] = -\frac{1}{4}[[X,Y],Z]. \end{aligned}$$

The second equality is Jacobi's identity. The metric invariance proved in Lecture 16 gives $g([Y,B],X) + g(B,[Y,X]) = 0$ for $B = [X,Y]$. Consequently,

$$g(R(X,Y)Y,X) = -\frac{1}{4}g([B,Y],X) = \frac{1}{4}g(B,B).$$

This is the sectional-curvature formula for the orthonormal pair. ∎

Example 17.11 (A bi-invariant metric on $SU(2)$)**.** We use the real inner product $g(A,B) = -2\operatorname{tr}(AB)$ on the traceless anti-Hermitian matrices $\mathfrak{su}(2)$. It is positive because $-\operatorname{tr}(A^2) = \operatorname{tr}(A^\dagger A)$. Trace invariance under conjugation makes its left-translated metric bi-invariant. An orthonormal basis is

$$E_1 = \frac{1}{2}\begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}, \quad E_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad E_3 = \frac{1}{2}\begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}.$$

Matrix multiplication gives $[E_1, E_2] = E_3$ and its cyclic counterparts. Thus $K(\operatorname{span}\{E_1, E_2\}) = \frac{1}{4}$. The positive value agrees with our sphere sign convention.

Self-check. State in one sentence what the curvature tensor is and why it was introduced. Explain why a nonzero Christoffel symbol does not by itself establish nonzero curvature.

Answer. The curvature tensor is the function-linear operator measuring the leading failure of parallel transport to return a vector unchanged around small coordinate loops. The polar plane has nonzero Christoffel symbols but $R^r{}_{\phi r\phi} = -1 + 1 = 0$. The derivative and product terms cancel, so a connection coefficient alone does not measure curvature.

Unit 17B Moving frames and surface curvature

17.4 Moving frames and Cartan's equations

The spherical coordinate vectors have varying lengths. An orthonormal frame separates these lengths from the rotation of the frame. Its connection forms record that rotation, while its coframe records the metric.

Definition 17.12. A *local frame* $e = (e_1, \dots, e_n)$ is a smooth basis of tangent vectors at every point of an open set U. It is a row of fields.

Definition 17.13. Its *dual coframe* is the column of 1-forms $\theta = (\theta^1, \dots, \theta^n)^T$ with $\theta^i(e_j) = \delta^i_j$.

Definition 17.14. For a Riemannian metric, a frame is *orthonormal* when $g(e_i, e_j) = \delta_{ij}$.

Gram–Schmidt applied smoothly to a coordinate frame supplies local orthonormal frames. The construction divides only by positive smooth lengths. The dual coframe then gives $g = \sum_i \theta^i \otimes \theta^i$. The figure compares coordinate vectors with unit frame vectors on a sphere.

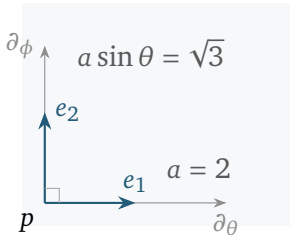


Figure 17.2: For a radius-a sphere, the spherical coordinate vectors have lengths a and $a\sin\theta$ (drawn at $a = 2$ and $\theta = \pi/3$). Dividing by those lengths gives the orthonormal frame e_1 and e_2 on a chart away from the poles.

Definition 17.15. The *connection forms* $\omega^i{}_j$ are specified by

$$\nabla_X e_j = e_i \omega^i{}_j(X).$$

Definition 17.16. The *curvature forms* $\Omega^i{}_j$ are specified by

$$R(X,Y)e_j = e_i \Omega^i{}_j(X,Y).$$

Linearity of the connection in its direction makes the first coefficients 1-forms. Tensoriality and direction antisymmetry make the second 2-forms. In an orthonormal frame, $\omega^i{}_j(X)$ is the component along e_i of the rate of change of e_j in direction X. The first structure equation below expresses zero torsion. The second computes curvature from those rates. Write ∇V for the vector-valued 1-form $X \mapsto \nabla_X V$. The matrices act on coefficient columns: for $V = ev$, $\nabla V = e(\mathrm{d}v + \omega v)$. For matrices of forms, the wedge includes matrix multiplication: $(A \wedge B)^i{}_j = \sum_k A^i{}_k \wedge B^k{}_j$. The exterior derivative acts on each entry.

Lemma 17.17 (Exterior derivative on fields)**.** *For a smooth 1-form α and smooth fields X and Y,*

$$(\mathrm{d}\alpha)(X,Y) = X(\alpha(Y)) - Y(\alpha(X)) - \alpha([X,Y]).$$

Proof. The right side is function-linear in X and Y by the product and bracket rules. On coordinate fields it agrees with Lecture 10's definition of $\mathrm{d}\alpha$. Tensoriality therefore gives equality for all fields. For example, when $\alpha = f\mathrm{d}g$, the right side is $X(f)Y(g) - Y(f)X(g)$, the evaluation of $\mathrm{d}f \wedge \mathrm{d}g$. ∎

Theorem 17.18 (Cartan's structure equations)**.** *On a Riemannian manifold, the Levi-Civita connection in a local orthonormal frame satisfies*

$$\mathrm{d}\theta = -\omega \wedge \theta, \qquad \Omega = \mathrm{d}\omega + \omega \wedge \omega.$$

The connection matrix ω and curvature matrix Ω are skew-symmetric. Conversely, a skew matrix ω satisfying the first equation is the unique Levi-Civita connection matrix for this coframe.

Proof. Differentiate $g(e_i, e_j) = \delta_{ij}$. Metric compatibility gives $\omega^i{}_j + \omega^j{}_i = 0$. For any connection matrix in this frame, the lemma gives

$$\theta^i(T(X,Y)) = (\mathrm{d}\theta^i + \omega^i{}_j \wedge \theta^j)(X,Y).$$

Indeed, expand $Y = \theta^j(Y)e_j$ in $\nabla_X Y$ and subtract the reversed expression and $[X,Y]$. Zero torsion therefore gives the first structure equation.

Conversely, a smooth matrix ω defines a connection by $\nabla_X(ev) = e(Xv + \omega(X)v)$. The product rule verifies both connection axioms. If ω is skew, the extra terms in differentiating $u^T v$ cancel because $u^T(\omega(X)^T + \omega(X))v = 0$. Thus this connection is metric. The displayed torsion identity makes it torsion-free when the first equation holds. Uniqueness in Theorem 16.6 identifies it with the Levi-Civita connection, proving the converse and uniqueness of the skew solution.

For the second, expand $\nabla_X \nabla_Y e_j$ using the Leibniz rule and subtract its reversed expression and $\nabla_{[X,Y]} e_j$. The coefficient of e_i is

$$X\omega^i{}_j(Y) - Y\omega^i{}_j(X) - \omega^i{}_j([X,Y]) + \omega^i{}_k(X)\omega^k{}_j(Y) - \omega^i{}_k(Y)\omega^k{}_j(X).$$

The first terms are $\mathrm{d}\omega^i{}_j(X,Y)$ and the last terms are $(\omega \wedge \omega)^i{}_j(X,Y)$. This proves the curvature equation. The last-pair metric antisymmetry already proved gives $\Omega^i{}_j = -\Omega^j{}_i$. ∎

17.5 Gaussian curvature of surfaces

Definition 17.19. On a Riemannian surface, the *Gaussian curvature* K is the sectional curvature of its tangent plane.

Choose a positively oriented orthonormal frame on a chart. Write its connection entry as $\omega^1{}_2$. Skew-symmetry makes the matrix

$$\omega = \begin{pmatrix} 0 & \omega^1{}_2 \\ -\omega^1{}_2 & 0 \end{pmatrix}.$$

Cartan's equations become

$$\mathrm{d}\theta^1 = -\omega^1{}_2 \wedge \theta^2, \qquad \mathrm{d}\theta^2 = \omega^1{}_2 \wedge \theta^1, \qquad \mathrm{d}\omega^1{}_2 = K\theta^1 \wedge \theta^2.$$

For the last equality, the diagonal-free 2×2 matrix has $(\omega \wedge \omega)^1{}_2 = 0$. The coefficient of $\Omega^1{}_2$ on (e_1, e_2) is $g(R(e_1, e_2)e_2, e_1) = K$. Thus the curvature coefficient has the displayed positive sign. In terms of the frame, the same convention is $\nabla e_1 = -\omega^1{}_2 e_2$ and $\nabla e_2 = \omega^1{}_2 e_1$.

Proposition 17.20 (Surface curvature)**.** *On a Riemannian surface, the lowered-output curvature satisfies*

$$R_{abcd} = K(g_{ac}g_{bd} - g_{ad}g_{bc}), \qquad \mathrm{Ric} = Kg, \qquad R_{\mathrm{sc}} = 2K.$$

The round sphere of radius $a > 0$ has $K = 1/a^2$. The Poincaré half-plane with metric $(\mathrm{d}x \otimes \mathrm{d}x + \mathrm{d}y \otimes \mathrm{d}y)/y^2$ has $K = -1$. The torus $\mathbb{R}^2/(2\pi\mathbb{Z})^2$ with its quotient Euclidean metric has $K = 0$.

In two dimensions, antisymmetry in both pairs leaves a single independent curvature component in an orthonormal basis. Its value is K. Tensoriality therefore gives the displayed formula in any coordinates. Contracting its first and third indices with the inverse metric gives $\mathrm{Ric} = Kg$ and $R_{\mathrm{sc}} = 2K$. This is intrinsic; no embedding is required.

Definition 17.21. For a surface embedded in $\mathbb{R}^3$ with a local unit normal n, its *principal curvatures* κ_1 and κ_2 at p are the eigenvalues of the shape operator $-\mathrm{d}n_p : T_pM \to T_pM$.

The normal sections in the principal directions have these signed curvatures. Tu's Theorem 12.3, p. 90, identifies intrinsic Gaussian curvature with $K = \kappa_1\kappa_2$.[2]

With outward normals, a unit sphere has both principal curvatures -1. A unit cylinder has principal curvatures -1 and 0. For the upward normal on $z = (x^2 - y^2)/2$, their values at the origin are 1 and -1. Their products give the signs shown in the figure.

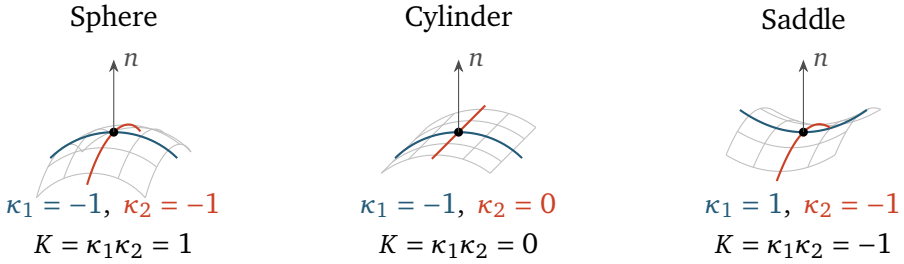


Figure 17.3: Normal sections in principal directions on a sphere, cylinder and saddle. With the indicated normals, their principal-curvature products are 1, 0 and −1.

[2] With our frame action, $\nabla e_1 = -\omega^1{}_2 e_2$ and $\nabla e_2 = \omega^1{}_2 e_1$. His (12.9) then reads $\mathrm{d}\omega^1{}_2 = +K\theta^1 \wedge \theta^2$.

The forms calculations for the sphere and half-plane follow in Examples 17.22 and 17.23. For the torus $\mathbb{R}^2/(2\pi\mathbb{Z})^2$, use the metric induced by the Euclidean plane under the quotient charts. Their transition maps are translations, so the metric agrees on overlaps. Its coordinate coefficients are constant, hence its Christoffel symbols and curvature vanish. The figure shows why the Cartesian frame also descends.

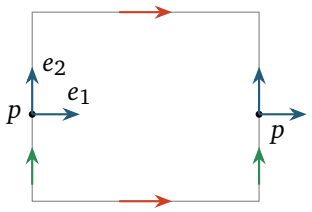


Figure 17.4: Opposite sides are identified by translations. The same Cartesian orthonormal frame matches across both pairs, giving the flat quotient metric.

This metric differs from the metric induced on a doughnut-shaped surface in $\mathbb{R}^3$.

17.6 The sphere and half-plane by forms

Example 17.22 (The sphere in an orthonormal frame)**.** We compute the curvature of the radius-a sphere of Example 17.5 using Cartan's equations. Here θ is the polar angle, while θ^1 and θ^2 are coframe entries, not powers of that angle. The coframe is

$$\theta^1 = a\mathrm{d}\theta, \qquad \theta^2 = a\sin\theta\mathrm{d}\phi.$$

Their derivatives are $\mathrm{d}\theta^1 = 0$ and $\mathrm{d}\theta^2 = a\cos\theta\mathrm{d}\theta\wedge\mathrm{d}\phi$. We solve for the connection coefficients by expanding in coordinate 1-forms. Write $\omega^1{}_2 = p\mathrm{d}\theta + q\mathrm{d}\phi$ for smooth functions p and q. The first structure equation gives

$$0 = -pa\sin\theta\mathrm{d}\theta\wedge\mathrm{d}\phi.$$

Since $a > 0$ and $0 < \theta < \pi$, this forces $p = 0$. The second equation then gives

$$a\cos\theta\mathrm{d}\theta\wedge\mathrm{d}\phi = q\mathrm{d}\phi\wedge a\mathrm{d}\theta = -qa\mathrm{d}\theta\wedge\mathrm{d}\phi.$$

Thus $q = -\cos\theta$, so the unique skew solution is

$$\omega^1{}_2 = -\cos\theta\mathrm{d}\phi.$$

It follows that

$$\mathrm{d}\omega^1{}_2 = \sin\theta\mathrm{d}\theta\wedge\mathrm{d}\phi = \frac{1}{a^2}\theta^1\wedge\theta^2.$$

Uniqueness of the Levi-Civita connection makes this the same curvature. Both routes give $\mathrm{Ric} = g/a^2$ and $R_{\mathrm{sc}} = 2/a^2$. These are smooth tensor identities on a dense coordinate domain, so continuity extends them, and $K = 1/a^2$, across the poles and angular seams.

A unit tangent vector at height y in the half-plane has Euclidean length y. The figure shows the corresponding tangent unit circles at several points.

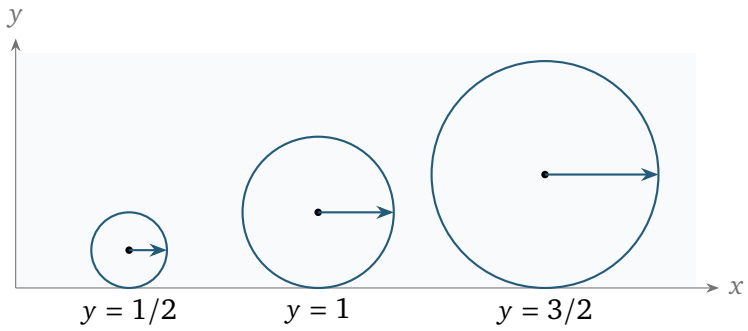


Figure 17.5: Each circle represents unit tangent vectors at its marked base point for $g = (\mathrm{d}x^2 + \mathrm{d}y^2)/y^2$; it is not a metric circle of points.

Example 17.23 (The Poincaré half-plane)**.** On the half-plane $y > 0$, choose $\theta^1 = \mathrm{d}x/y$ and $\theta^2 = \mathrm{d}y/y$. The derivatives and solution of the first equation are

$$\mathrm{d}\theta^1 = \frac{\mathrm{d}x \wedge \mathrm{d}y}{y^2}, \qquad \mathrm{d}\theta^2 = 0, \qquad \omega^1{}_2 = -\frac{\mathrm{d}x}{y}.$$

Taking its derivative gives $\mathrm{d}\omega^1{}_2 = -\mathrm{d}x \wedge \mathrm{d}y/y^2 = -\theta^1 \wedge \theta^2$, so $K = -1$.

The flat quotient torus above has a different metric from an embedded torus. For an embedded torus with major radius b and tube radius r, we use $b > r > 0$ and the parametrization

$$p(u, v) = ((b + r\cos v)\cos u, (b + r\cos v)\sin u, r\sin v).$$

The figure locates the regions of positive and negative curvature.

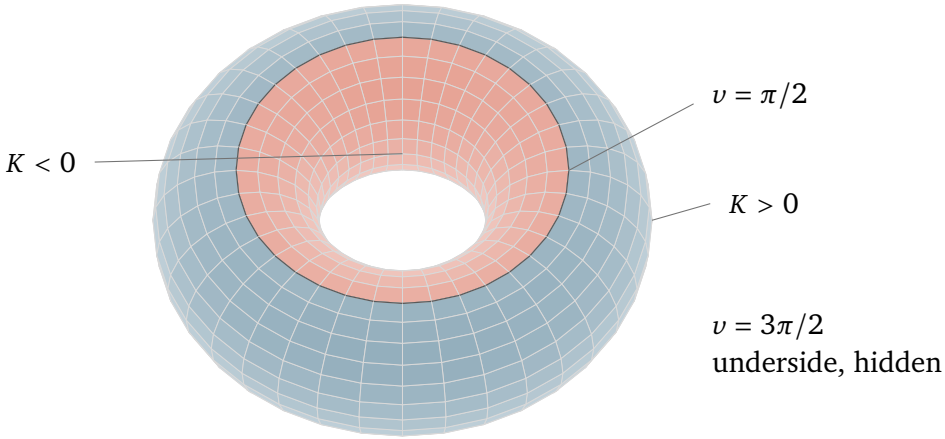


Figure 17.6: For $b = 2$ and $r = 1$, the outer region $\cos v > 0$ has positive curvature, and the inner region $\cos v < 0$ has negative curvature; the circles $v = \pi/2$ and $v = 3\pi/2$ have zero curvature.

Example 17.24 (An embedded torus)**.** Differentiating the parametrization gives the metric and coframe

$$g = r^2\mathrm{d}v \otimes \mathrm{d}v + (b + r\cos v)^2\mathrm{d}u \otimes \mathrm{d}u, \qquad \theta^1 = r\mathrm{d}v, \quad \theta^2 = (b + r\cos v)\mathrm{d}u.$$

We choose the chart orientation (v, u), so $\theta^1 \wedge \theta^2$ is positive. The derivatives are $\mathrm{d}\theta^1 = 0$ and $\mathrm{d}\theta^2 = -r\sin v\mathrm{d}v \wedge \mathrm{d}u$. Writing $\omega^1{}_2 = p\mathrm{d}v + q\mathrm{d}u$ in the first equation gives $p = 0$. The

second gives $-r\sin v = -rq$, hence $q = \sin v$. Therefore

$$\omega^1{}_2 = \sin v\,\mathrm{d}u, \qquad \mathrm{d}\omega^1{}_2 = \cos v\,\mathrm{d}v \wedge \mathrm{d}u, \qquad K = \frac{\cos v}{r(b + r\cos v)}.$$

The denominator is positive because $b > r > 0$. Thus the curvature is positive outside, negative inside and zero on the two marked circles.

The second Bianchi identity

Curvature forms satisfy a differential identity because $\mathrm{d}^2 = 0$. Lecture 19 contracts this identity to obtain a divergence-free tensor in the gravitational field equation. The curvature-form calculation used no metric hypothesis. Thus $\Omega = \mathrm{d}\omega + \omega \wedge \omega$ holds in any local frame for any affine connection. Differentiate a tensor by the connection product rule: $\nabla_X T$ differentiates the output and subtracts a term for the covariant derivative of each input. For curvature, this means

$$\begin{aligned}(\nabla_X R)(Y,Z)W = {} & \nabla_X(R(Y,Z)W) - R(\nabla_X Y, Z)W \\ & - R(Y, \nabla_X Z)W - R(Y,Z)\nabla_X W.\end{aligned}$$

Proposition 17.25 (Second Bianchi identity · *sketched; matrix-form identity proved*)**.** *For the Levi-Civita connection of a pseudo-Riemannian metric,*

$$\mathrm{d}\Omega + \omega \wedge \Omega - \Omega \wedge \omega = 0, \qquad \sum_{\mathrm{cyc}(X,Y,Z)} (\nabla_X R)(Y,Z) = 0.$$

Proof and sketch. The identity $\mathrm{d}^2\omega = 0$ gives $\mathrm{d}\Omega = \mathrm{d}\omega\wedge\omega - \omega\wedge\mathrm{d}\omega$. Substitute $\mathrm{d}\omega = \Omega - \omega\wedge\omega$; the cubic terms cancel. This proves the matrix equation.[3] Evaluating it on three directions and using zero torsion cancels their bracket terms, giving the stated cyclic covariant identity. This sketches the conversion from forms to the covariant expression. ∎

The cylinder's angular chart also gives coordinates on a Euclidean strip. The figure displays the local length-preserving map used in the self-check.

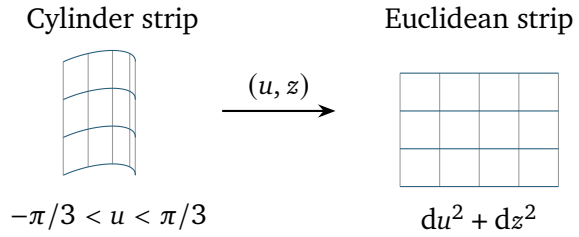


Figure 17.7: An open angular strip on the unit cylinder maps to a Euclidean strip by $(u, z) \mapsto (u, z)$; for radius a, the horizontal coordinate is au.

Self-check. Compute the intrinsic Gaussian curvature of the cylinder of radius $a > 0$.

Answer. The parametrization $p(u, z) = (a\cos u, a\sin u, z)$ induces $g = a^2\mathrm{d}u \otimes \mathrm{d}u + \mathrm{d}z \otimes \mathrm{d}z$. Use $\theta^1 = a\mathrm{d}u$ and $\theta^2 = \mathrm{d}z$ on an angular chart. Both forms are closed, so the skew connection matrix $\omega = 0$ satisfies Cartan's first equation and is the matrix of the unique metric torsion-free connection. Its curvature is zero, hence $K = 0$. The coordinates (au, z) make the metric

[3] Nakahara's (7.147b) states the matrix identity with the same action convention.

Euclidean on each angular chart. This is a local isometry, not a global identification of the cylinder with a plane. The cylinder bends in its embedding but has zero intrinsic curvature.

Self-check. On $\mathbb{R}^2$ with coordinates (x, y) and metric $g = \mathrm{d}x \otimes \mathrm{d}x + (1+x^2)^2 \mathrm{d}y \otimes \mathrm{d}y$, solve for $\omega^1{}_2$ in the coframe $\theta^1 = \mathrm{d}x$ and $\theta^2 = (1+x^2)\mathrm{d}y$. Find K. Repeat for the polar plane with $\theta^1 = \mathrm{d}r$ and $\theta^2 = r\mathrm{d}\phi$, $r > 0$.

Answer. For a coframe $\mathrm{d}x$, $f(x)\mathrm{d}y$ with $f > 0$, write $\omega^1{}_2 = p\mathrm{d}x + q\mathrm{d}y$. The first equation gives $pf = 0$, so $p = 0$. The second gives $f' = -q$. Thus $\omega^1{}_2 = -f'\mathrm{d}y$ and $\mathrm{d}\omega^1{}_2 = -f''\mathrm{d}x \wedge \mathrm{d}y$, so $K = -f''/f$. For $f(x) = 1 + x^2$, these formulas give

$$\omega^1{}_2 = -2x\mathrm{d}y, \qquad K = -\frac{2}{1+x^2}.$$

For the polar plane, $f(r) = r$ gives $\omega^1{}_2 = -\mathrm{d}\phi$ and $K = 0$. A nonzero connection form can therefore have zero curvature.

Summary. Curvature is the tensor $R(X,Y) = [\nabla_X, \nabla_Y] - \nabla_{[X,Y]}$. Metric compatibility and zero torsion give its antisymmetries and first Bianchi identity. For a Riemannian metric, Ricci sums sectional curvatures through a unit direction, and scalar curvature is twice the sum over basis planes. A bi-invariant metric gives $R(X,Y)Z = -\frac{1}{4}[[X,Y],Z]$ on left-invariant fields. In a local orthonormal frame, Cartan's equations replace Christoffel computations by $\mathrm{d}\theta = -\omega \wedge \theta$ and $\Omega = \mathrm{d}\omega + \omega \wedge \omega$. On a surface, $\mathrm{d}\omega^1{}_2 = K\theta^1 \wedge \theta^2$. The sphere gives positive curvature, the half-plane negative curvature, and the flat quotient torus and cylinder zero curvature. Lecture 18 integrates the surface equation to compute parallel transport around loops.

Reading. Tu, *Differential Geometry*, §10–§12, especially Theorems 11.1 and 11.7 and Theorem 12.3, p. 90. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Chapter 7, Propositions 7.1, 7.4 and 7.5. Nakahara, §7.3 and §7.8.